The Apache Spark Ecosystem

There are different technologies that you might come across, when you start working with spark and pyspark. There are lots of different components with similar names, so we are going to make sure we can navigate through them.

1. **Spark Core:** The foundation of spark is called spark core, which provides distributed task dispatching, scheduling, and basic input/output functionalities.

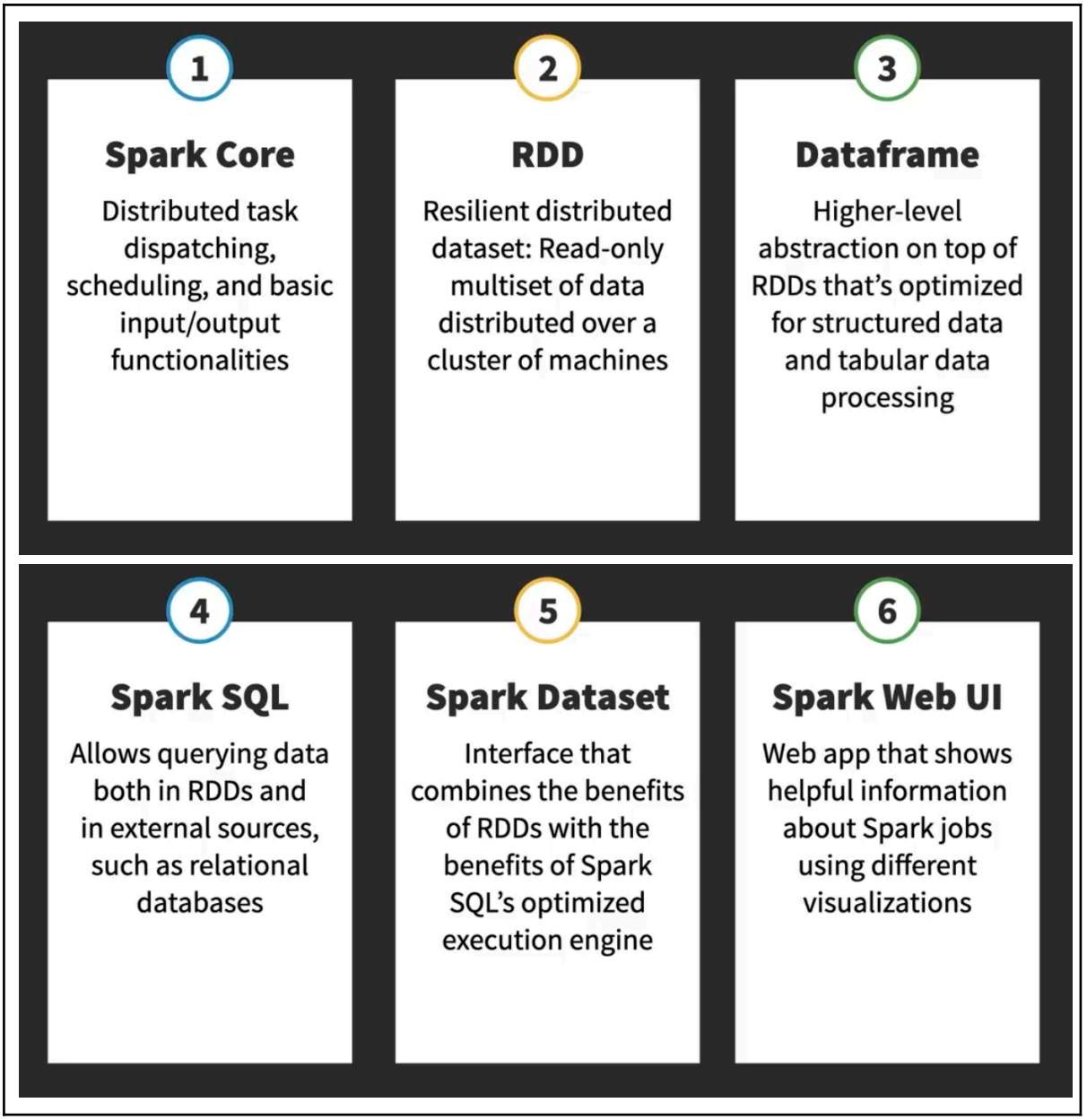
2. **Resilient Distributed Dataset (RDD)**: A component of spark which is a read-only multiset of data items that's distributed over a cluster of machines and maintained in a fault-tolerant way.

3. **Dataframe:** A dataframe is a higher level abstraction on top of the RDDs, that's optimized for structured data and tabular data processing. It is conceptually the same as a table in a relational database or a dataframe in python.

4. **Spark SQL:** Spark sql is a spark module that allows us to query data both in RDDs and in external sources, such as relational databases.

5. **Spark Dataset:** The spark dataset is a new interface that is only available through the scala and Java API to spark. A dataset combines both the benefits of RDDs such as strong typing, with the benefit of Spark SQL's optimized execution engine. You can also think of a dataframe as a dataset, that's organized into named columns.

6. **Spark Web UI:** The spark Web UI is a web app that comes bundled with spark and can be accessed through our web browser. This UI shows helpful information about the spark jobs that you're running, so you can monitor them.



Finally, there are 2 more components of the spark ecosystem that you might come across:

- **Structured Streaming:** It is a scalable and fault-tolerant stream processing engine built on the spark sql engine.

- **Machine Learning Library:** It is a library that contains machine learning algorithms and other functionality that makes practical machine learning in spark scalable and easy.